

Business Requirements Document

Antenna Recommender (ANT)

Document ID ANT-INIT-BRD	Phase Initiating	Version 1.0
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1. Executive Summary

This Business Requirements Document defines the business problem, goals, scope, and stakeholders for the Antenna Recommender project — a website that helps students and Communication engineers select the most suitable antenna for their technical requirements. It serves as the foundation from which the Product Requirements Document (PRD) and subsequent technical design are derived.

2. Business Problem

Students and RF engineers often struggle to choose the appropriate antenna because antenna selection depends on several interacting technical parameters. Existing resources are scattered across multiple sources and require a level of technical expertise that is a barrier for students and non-specialist engineers, slowing down design and learning workflows.

3. Business Goal

Develop a website that simplifies antenna selection by recommending the most suitable antenna based on user-entered requirements — reducing reliance on scattered, expertise-dependent reference material.

4. Stakeholders

Stakeholder	Responsibility
Project Sponsor (Dr. Randa Fouad)	Approves the project and provides overall guidance; provides high-level project funding and signs acceptance of final deliverables.
Project Manager (Nourin Elshenawi)	Plans, coordinates, and manages the project; authorized to manage the team, execute the plan, allocate resources, and make decisions to keep the project on track.
Project Team	Designs, develops, and tests the application.
Subject Matter Experts (SMEs)	Verify the technical accuracy of antenna recommendations.
End Users	Use the application and evaluate its usability.

4.1 Stakeholder Analysis

Stakeholder	Power	Interest	Engagement Strategy
Project Sponsor	High	High	Manage Closely — keep fully informed, consult before major decisions (e.g. scope change), weekly status reports.
Project Manager	High	High	Manage Closely — maintain full oversight, allocate resources, decide on scope, time, and cost impacts.
Project Team	Medium	High	Keep Informed — involve in planning (WBS), assign clear tasks (RACI), maintain daily coordination.
SMEs	Medium	Medium	Keep Informed — consult during the technical requirements phase; secure final approval on algorithm accuracy.
End Users	Low	Medium	Keep Satisfied — gather initial requirements (UI/UX, features); validate final usability.

5. Business Scope

5.1 In Scope

- The website.
- The recommendation algorithm.
- The user interface.
- The antenna comparison feature.
- Project documentation.

5.2 Out of Scope

- Physical antenna testing.
- Real-time external databases.
- User accounts and login.

6. Business Requirements

The following business-level requirements translate the business goal into conditions the solution must satisfy. Detailed functional and non-functional specifications are defined in the companion Product Requirements Document (PRD).

ID	Business Requirement	Rationale
BR-01	The system shall allow users to enter technical parameters (frequency, application type, required gain).	Directly addresses the business problem of scattered, hard-to-use reference material.
BR-02	The system shall recommend the most suitable antenna along with alternative options.	Reduces reliance on specialist expertise for antenna selection.
BR-03	The system shall provide a comparison of key specifications across recommended options.	Supports informed decision-making by the end user.

ID	Business Requirement	Rationale
BR-04	The system shall be accessible as a responsive website with no login required.	Keeps the tool frictionless and available to students and engineers without account overhead.
BR-05	Recommendation logic shall be validated by Subject Matter Experts.	Ensures technical credibility and accuracy of results.

7. Assumptions & Constraints

7.1 Assumptions

- Antenna reference data can be sourced from published technical data and validated by SMEs.
- Users have basic familiarity with RF terminology (frequency, gain, application type).
- The project team has continued access to SME support throughout the technical requirements phase.

7.2 Constraints

- No physical antenna testing or fabrication is within scope.
- No real-time external database integration is within scope.
- The project is delivered on an academic timeline with a 7-week high-level schedule.

8. Risks

Risk	Description
Technical data risk	Difficulty in finding reliable, validated datasets for the antenna recommendation algorithm.
Resource availability risk	Potential scheduling conflicts among team members due to concurrent academic commitments.
Scope creep	Late requests for complex, specialized recommendation features not initially budgeted.

9. Success Criteria / Acceptance

- All planned activities (including PM documentation and the final web application) are completed and accepted by the Sponsor.
- Schedule Performance Index (SPI) ≥ 1.0 at final project closure.
- The web application successfully recommends at least two distinct antenna types based on a user-defined frequency and application type.
- Formal acceptance of the final deliverable by the Sponsor.

10. High-Level Milestones

Phase	Key Deliverables
Initiating	Project Charter, Stakeholder Register, Naming Convention Plan.
Planning	Scope Statement, WBS & Dictionary, Schedule Baseline, Cost & Quality Plans, Risk Register.
Executing & Monitoring/Controlling	Status Reports, Change Requests, Performance Dashboard.
Closing	Final Project Report.
Product	The fully functional Antenna Recommender (front-end and back-end code).